



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ELECTRONICS & COMMUNICATION ENGINEERING

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 EC 3101	Linear ICs & Pulse Circuits	PC	3	3	--	--	25	75	100
B19 EC 3102	Digital Communication	PC	3	3	--	--	25	75	100
B19 EC 3103	Antennas & Propagation	PC	3	3	--	--	25	75	100
B19 EC 3104	Control Systems	PC	3	3	--	--	25	75	100
B19 EC 3105	Electronic Measurements & Instrumentation	PC	3	3	--	--	25	75	100
# PE-I	Professional Elective-I	PE	3	3	--	--	25	75	100
B19 EC 3110	Digital Communication Lab - (with Simulation)	PC	1.5	--	--	3	20	30	50
B19 EC 3111	Linear Integrated Circuits & Pulse Circuits - Lab (with Simulation)	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability Skills-I	MC	--	3	--	--	--	--	--
B19 MC 3102	Basic Coding	MC	--	--	--	3	--	--	--
TOTAL			21	21	--	9	190	510	700

#PE-I	Course Code	Course
	B19 EC 3106	Data Communications & Computer Networks (DCCN)
	B19 EC 3107	Digital System Design using HDL
	B19 EC 3108	Soft Computing Techniques
	B19 EC 3109	Operating systems

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3101	PC	3	--	--	3	25	75	3 Hrs.
LINEAR IC'S AND PULSE CIRCUITS								
(For ECE)								
Course Objectives:								
1.	To analyze linear and non-linear wave shaping circuits							
2.	To design various multivibrators for various applications							
3.	To design linear and non-linear applications of operational amplifier							
4.	To analyze the concepts of Active filters and waveform generators							
5.	To design the basic applications using 555 timers							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							K L
1.	Analyze passive RC circuits, clippers, and clamper circuits.							K4
2.	Design different multivibrators for various applications							K4
3.	Design/analyze fundamental circuits based on op-amps							K4
4.	Design and analyze of various active filters, oscillators							K4
5.	Design and analyze of various applications using IC 555 timer, IC565 PLL.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Wave Shaping: Response of High pass and Low pass RC circuits to step, square inputs. High pass RC circuit as a differentiator, Low pass RC circuit as an integrator. Diode clippers, clipping at two independent levels, Clamping Operation, Clamping Circuits with Different Inputs, Clamping Circuit Theorem							
UNIT-II (10 Hrs)	Multivibrators: Transistor as a switch, switching times of a transistor, Design and Analysis of Fixed-bias Bistable multivibrator, Design and analysis of Collector coupled Monostable Multivibrator, Expression for the gate width and its waveforms, Design and analysis of Collector coupled Astable Multivibrator, Expression for the Time period and its waveforms, The Astable Multivibrator as a voltage to frequency converter							
UNIT-III (12 Hrs)	Applications of Operational Amplifiers: Basics of Op-Amp, Block Diagram, Ideal characteristics, op-amp parameters, open loop, and closed loop op-am configurations, Logarithmic amplifier, Instrumentation Amplifier, Op-amp as a Comparator, Schmitt trigger, square wave generator, Triangular wave generator, Rectifiers.							
UNIT-IV (9 Hrs)	Active Filters and Oscillators: Introduction to active filters, Butterworth type LPF, HPF first and second order filters, Higher order filters, Wide Band pass filters, Wide band reject filters, All pass filters. Op-Amp Phase Shift oscillator, Wein-bridge oscillator, Quadrature Oscillator, voltage-controlled oscillators, Voltage to current and current to voltage converters							
UNIT-V (10 Hrs)	Special ICs: 555 Timers, Introduction, Block diagram, 555 timer as an Astable and Monostable Multivibrator, Applications of 555 Timer as Monostable multivibrator- Frequency divider,							

	Pulse stretcher, Missing pulse detector, Pulse width modulation, Applications of 555 Timer as Astable Multivibrator- Square wave oscillator, FSK Generator, Schmitt trigger, IC 565 PLL Block diagram, Monolithic IC565 PLL, Applications- Frequency multiplication/division, Frequency translation, Frequency adder/subtractor
Text Books:	
1.	Pulse Digital and Switching Waveforms, J. Millman,H. Taub, and M.S.PrakashRao McGraw-Hill, Second Edition
2.	Linear Integrated Circuits- D.Roy Chowdhury, New Age International(p) Ltd, 2nd Edition,2003
3.	Op-Amps and Linear ICs- RamakanthGayakwad, PHI, 1987.
Reference Books:	
1.	Integrated Circuits- Botkar, Khanna Publications
2.	Applications of Linear ICs- Clayton
3.	Pulse and Digital Circuits, A. Anand Kumar, PHI, Second Edition, 2005
Web links	
1.	<u>Integrated Circuits, MOSFETs, Op-Amps and their Applications</u> https://nptel.ac.in/courses/108/108/108108111/
2.	<u>Electronic Modules for Industrial Applications using Op-Amps</u> https://nptel.ac.in/courses/108/108/108108125/
3.	<u>Op-Amp Practical Applications: Design, Simulation and Implementation</u> https://nptel.ac.in/courses/108/108/108108114/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3102	PC	3	--	--	3	25	75	3 Hrs.
DIGITAL COMMUNICATION								
(For ECE)								
Course Objectives:								
1.	To introduce the elementary concepts of digital communication systems.							
2.	To get introduced with emphasis on different modulation techniques.							
3.	Understand the effect of noise on signal transmission.							
4.	To learn about optimum detection and probability of error.							
5.	To compare the performance of two digital modulation techniques and introduce the elementary concept of spread spectrum modulation system.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							K L
1.	Understand the basic concepts of sampling and digital communication systems.							K2
2.	Understand the concept of binary and M-Ary modulation techniques.							K2
3.	Apply the knowledge of signals & systems and evaluate the performance of various filters in the presence of noise.							K3
4.	Understanding the concept of probability of error & by applying the knowledge of basic digital modulation techniques to evaluate their optimal performance.							K3
5.	Analyze the error performance of two digital modulation techniques and understand the concept of spread spectrum communication system							K3
SYLLABUS								
UNIT-I (12 Hrs)	Pulse Modulation and Digital Representation of Analog Signal: Sampling, Cross Talk, Digital representation of analog signal: Quantization of signals, Quantization error, Pulse Code Modulation, Companding, T1 Digital system, Differential Pulse Code Modulation, Delta Modulation, Adaptive Delta Modulation, Continuously Variable Slope Delta Modulation.							
UNIT-II (12 Hrs)	Digital Modulation and Transmission Binary Phase-Shift Keying, Differential Phase-Shift Keying, Differentially-Encoded PSK(DEPSK), Quadrature Phase-Shift Keying (QPSK), M-ary PSK, Amplitude Shift Keying, Binary Frequency Shift-Keying, Comparison of BFSK and BPSK, M-ary FSK, Minimum Shift Keying (MSK).							
UNIT-III (8 Hrs)	Mathematical Representation of Noise: Some Sources of Noise, Frequency-domain representation of Noise, Spectral Components of Noise, Response of a Narrowband Filter to Noise, Effect of a Filter on the Power Spectral Density of Noise, Superposition of Noises, Linear Filtering, Noise Bandwidth, Quadrature Components of Noise, Power Spectral Density of Quadrature Components of Noise.							

UNIT-IV (8 Hrs)	Optimal Reception of Digital Signal: A Base-band Signal Receiver, Probability of Error, Optimum Receiver for both Baseband and Pass band - Calculation of optimum filter Transfer function, Optimum filter realization using Matched filter, Probability of Error of the Matched Filter, Optimum filter realization using Correlator, Optimal of Coherent Reception: PSK, FSK, QPSK, Comparison of Modulation Systems.
UNIT-V (12 Hrs)	a) Noise in Pulse Code Modulation and Delta Modulation Systems: PCM Transmission, Calculation of Signal-to-Noise Ratio in PCM, Delta Modulation (DM) Transmission, Calculation of Signal-to-Noise Ratio in DM, Comparison of PCM and DM. b) Introduction to Spread Spectrum Modulation: Direct Sequence (DS) Spread Spectrum, Use of Spread Spectrum with Code Division Multiple Access (CDMA), Ranging using DS Spread Spectrum, frequency hopping (FH), Characteristics of PN Sequence.
Text Books:	
1.	Principles of Communication Systems by Herbert Taub, Donald L Schilling and Goutam Saha, 3rd edition, Tata McGraw-Hill Publications, 2008 New Delhi.
2.	Digital Communications by Simon Haykins John Wiley, 2005
Reference Books:	
1.	Principles of Digital Communications- J.Das, SK.Mullick, P.K.Chatterjee.
2.	Modern Analog and Digital Communications by B.P.Lathi, Oxford reprint, 3rd Edition, 2004.
e-Resources:	
1.	https://classroom.google.com/c/NDA4Mjg4MjIxMzJa
2.	Electronics & Communication Engineering - Digital Communication - Prof.Bikas.Kumar.Dey, IIT Bombay.
3.	https://www.tutorialspoint.com/digital_communication/index.htm
4.	https://nptel.ac.in/courses/108/104/108104098/ -Prof .Aditya.K.Jagannatham, IIT Kanpur.

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EC3103	PC	3	--	--	3	25	75	3 Hrs.	
ANTENNAS AND PROPAGATION									
(For ECE)									
Course Objectives:									
1.	Understand the radiation mechanism of antennas and to learn about basic parameters like impedance, gain, directivity, bandwidth, effective length, beam width and radiation pattern etc.								
2.	Derive fields and power radiated by elemental antenna, Half wave dipole, quarter wave monopole and values of their radiation resistance.								
3.	Understand the necessity of antenna arrays and to learn about theory of uniform linear arrays, broad side and end fire arrays, non-uniform linear arrays like binomial arrays and pattern multiplication.								
4.	Acquire knowledge about practical LF, HF, VHF, UHF and Microwave antennas and be able to design practical antennas.								
5.	Acquire knowledge about various antenna measurements and be able to conduct different types of antenna measurements.								
6.	Gain knowledge about various types of radio wave propagation like Ground wave, Ionospheric, space wave and Duct propagation and be able to design different types of communication links.								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								K L
1.	Understand Radiation mechanism and functions of antennas identify antenna parameters and derive expressions for antenna parameters.								K3
2.	Analyze and design wire and aperture antennas for different applications.								K4
3.	Analyze and design Antenna arrays.								K4
4.	Capable of performing various antenna measurements and come up with conclusions about antenna parameters and performance.								K4
5.	Identify characteristics of radio wave propagation and be able to design different types of communication links for different frequency bands.								K4
SYLLABUS									
UNIT-I (8 Hrs)	Fundamentals of Antennas & Radiation from Antennas Functions and properties of antennas, antenna parameters, basic antenna elements, radiation mechanism, radiating fields of alternating current element, radiated power and radiation resistance of current element, different types of current distribution on linear antennas, radiated fields, radiated power and radiation resistance of half-wave dipole and quarter – wave monopole, directional characteristics of dipole antennas, transmission loss between transmitting and receiving antennas – Friis formula, antenna noise temperature and signal-to-noise ratio.								
UNIT-II (8 Hrs)	Linear Arrays: Uniform linear arrays, field strength of a uniform linear arrays, locations of principal maximum, nulls and secondary maxima, first side lobe level, analysis of broadside and end-fire arrays, Pattern multiplication, binomial arrays, effect of earth on vertical patterns Antenna array synthesis – Woodward Lawson method Dolphchebycheys linear array method.								

UNIT-III (15 Hrs)	Practical Antennas Classification of antennas according to type of radiation and type of current distribution of antennas – Isotropic, Omnidirectional & directional antennas, standing wave and travelling wave antennas, Classification according to frequency of operation – LF, MF, HF, VHF & UHF HF, VHF & UHF Antennas - V Antennas, Inverted V Antennas, Rhombic antennas, folded dipole, Yagi-Uda antenna, Log periodic antenna, Loop and Helical Antennas. Microwave antennas: Introduction, types of reflector antennas, corner reflector, parabolic reflector, feed systems for parabolic reflector, horn antennas, slot antennas and impedance of slot antennas, Babine's principle and micro strip antennas.
UNIT-IV (5 Hrs)	Antenna measurements Introduction, measurement ranges, antenna impedance measurements, antenna gain and directivity measurement, measurement of radiation pattern, beam width and SLL, Measurements of Polarization, Measurements of Phase, Measurements of Radiation Resistance
UNIT-V (10 Hrs)	Wave propagation Types of radio wave propagation, ground wave propagation and Sommerfeld's analysis of ground wave propagation, wave tilt of ground wave, structure of ionosphere, refractive index of ionosphere, mechanism of wave bending by ionosphere, critical frequency, MUF, Skip distance, fading and remedial measures, effect of earth's magnetic field on ionosphere propagation, Faraday rotation, tropospheric (space wave) propagation, range of space wave propagation, effective earth radius, field strength of space wave, atmospheric effects on space wave propagation, duct propagation and scatter propagation.
Text Books:	
1.	EM waves and Radiating systems – by E. C. JORDAN and K. G. Balmain – PHI , New Delhi.
2.	Antenna theory - by C. A. Balanis, John Wiley.
Reference Books:	
1.	Antennas and Wave Propagation – By J.D. Kraus, McGrawhill.
2.	Antennas and wave propagation – by G.S.N Raju, Pearson education.
3	Antenna and wave propagation – by K.D.Prasad
e-Resources:	
1.	https://www.tutorialspoint.com/antenna_theory/antenna_theory_types_of_propagation.htm
2.	https://youtu.be/wx_tIvaajAI__

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3104	PC	3	--	--	3	25	75	3 Hrs.
CONTROL SYSTEMS								
(For ECE)								
Course Objectives:								
1.	To learn the modeling of linear systems using transfer functions and obtain transfer functions from block diagram and signal flow graph representations.							
2.	To learn the significance of time response and find it for system analysis in transient and steady state.							
3.	To learn the concept of stability and know different techniques of stability analysis.							
4.	To learn the concept of frequency response and its application for control system analysis.							
5.	To learn the different kind of representation of a system with multi-input and multi output and analyze them.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Students will be able to represent mathematical models of systems using block diagrams & Signal Flow Graphs and derive their transfer functions.							K3
2.	Students will be able to analyze systems in time domain for transient and steady-state behavior.							K3
3.	Students will learn the concept of stability and use RH criterion and Root locus methods for stability analysis.							K3
4.	Students will learn to obtain frequency response plots of systems and use them for system analysis and stability assessment							K3
5.	Students will be able to analyze the system with multi-input and multi output							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to control systems: Introduction to control systems, Open loop and closed loop systems, Transfer Function and Impulse Response of Linear Systems, Block Diagrams of Control Systems and Reduction Techniques, Signal Flow Graph Representation of Control Systems, Mason's gain formula, Feedback Characteristics of Control Systems.							
UNIT-II (10 Hrs)	Time Domain Analysis of Control Systems: Time Response of First and Second Order Systems with Standard Input Signals, Time Domain Specifications of Second Order Systems, Steady State Error, Steady State Error Constants, Controllers: Proportional, Integral and Derivative Controllers, Proportional plus Integral, Proportional and Derivative, Proportional, Integral and Derivative Controllers.							
UNIT-III (8 Hrs)	Stability Analysis of Control Systems: Concept of Stability, Routh-Hurwitz Criterion, Relative Stability Analysis, The Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus							

UNIT-IV (10 Hrs)	Frequency Domain Analysis of control systems: Frequency Domain Specifications, Bode Plots- Log Magnitude versus Phase Plots, Polar Plots -Correlation between Time and Frequency Responses, Nyquist Stability Criterion - Assessment of Relative Stability, All Pass and Minimum Phase Systems, Gain Margin and Phase Margin Calculations for all Plots.
UNIT-V (8 Hrs)	State Space Analysis: Introduction to State Space Analysis, State Space Equation, State Model for Different Types of System Representations: Differential Equations, Electrical Network, Signal Flow Graph and Transfer Functions, Transfer Function form State Model, Controllability and Observability, Solution to the State Equation, State Transition Matrix and Properties.
Text Books:	
1.	I. J. Nagrath and M. Gopal, „ Control Systems Engineering“, New Age International Publishers (6 th Edition).
2.	Benjamin C. Kuo, „ Automatic Control Systems“, PHI (5th Edition).
Reference Books:	
1.	Katsuhiko Ogata, „Modern Control Engineering“, , PHI (4 th Edition).
2.	Richard C. Dorf and Robert H. Bishop, „Modern Control Systems“, Addison-Wesley Publishers (8 th Edition)

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3105	PC	3	--	--	3	25	75	3 Hrs.
ELECTRONIC MEASUREMENTS AND INSTRUMENTATION								
(For ECE)								
Course Objectives:								
1.	Select the instrument to be used based on the requirements.							
2.	Understand and analyze different signal generators and analyzers							
3.	Understand the design of oscilloscopes for different applications.							
4.	Understand the principle of operation and working of various types of bridges for measurement of parameters							
5.	Understand and analyze the different types of transducers							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Evaluate basics of measurement systems, principle of basic meter							K2
2.	Evaluate how a signal can be generated using different types of meters.							K3
3.	Investigate a signal / waveform with different oscillators.							K2
4.	Use bridges of many types and measure appropriate parameters.							K3
5.	Design different transducers for measurement of different parameters.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Qualities of Measurements: Introduction, Measurement standards, Performance characteristics of instruments, Static characteristics, Accuracy, Resolution, Precision, expected value, Error, Sensitivity. Errors in Measurement, Types of static errors-Gross errors, systematic errors, Instrumental errors, Observational errors, Random errors, Sources of error, Statistical analysis, Dynamic Characteristics-speed of response, Fidelity, Lag and Dynamic error. DC Voltmeters, Multi range, Range extension voltmeters, AC voltmeters, True RMS responding voltmeter, Electronic Multimeter.							
UNIT-II (8 Hrs)	Transducers: Active & passive transducers: Resistance, Capacitance, inductance; Resistive Transducer, Unbounded resistance wire Strain gauge, bonded resistance wire strain gauge, Semiconductor strain gauge, Linear Variable Differential Transducer, Piezo electric transducers, Resistance Thermometers, Thermocouples, Thermistors, Sensistors. Introduction to smart sensors.							
UNIT-III (8 Hrs)	Oscilloscopes: CRT features, Block diagram of oscilloscope, vertical amplifier, horizontal deflection system, sweep, trigger pulse, delay line. Dual beam CRO, Dual trace oscilloscope, sampling oscilloscope, digital storage oscilloscope, Lissajous method of frequency measurement, standard specifications of CRO, CRO probes. Introduction to Lab VIEW (Laboratory Virtual Instrument Engineering Workbench) and its advantages and features.							

UNIT-IV (8 Hrs)	Bridges: Measurement of inductance- Maxwell's bridge, Anderson bridge. Measurement of capacitance-Schering Bridge. Wheatstone bridge. Wien Bridge, Errors and precautions in using bridges.
UNIT-V (8 Hrs)	Signal Generator: Introduction, fixed frequency AF oscillator, variable frequency AF oscillator, Basic Standard signal generator, AF sine and square wave signal generators, Function Generators, Square pulse, Random noise, sweep, Arbitrary waveform. Wave Analyzers, Harmonic Distortion Analyzers, Spectrum Analyzers.
Text Books:	
1.	Electronic instrumentation, second edition - H.S.Kalsi, Tata McGraw Hill, 2004.
2.	Modern Electronic Instrumentation and Measurement Techniques – A.D. Helfrick and, D.W. Cooper, PHI, 5th Edition, 2002.
Reference Books:	
1.	Electronic Instrumentation & Measurements - David A. Bell, PHI, 2nd Edition, 2003.
2.	Electronic Test Instruments, Analog and Digital Measurements - Robert A. Witte, Pearson Education, 2nd Ed., 2004.
e-Resources:	
1.	Introduction to lab view (Laboratory Virtual Instrument Engineering Workbench) and its advantages and features. weblink: https://mndmajix.com/labview/introduction-to-labview-programming-and-its-advantages .

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3106	PE	3	--	--	3	25	75	3 Hrs.
DATA COMMUNICATIONS AND COMPUTER NETWORKS								
(For ECE)								
Course Objectives:								
1.	To introduce the Fundamentals of data communication networks							
2.	To familiarize with the fundamental concepts of computer networking and network engineering reference models.							
3.	To introduce basic concepts of multiplexing techniques, switching techniques.							
4.	To understand error control and flow control mechanisms.							
5.	To familiarize with different multiple access protocols such as ALOHA, CSMA.							
6.	To familiarize with different networking devices and congestion control algorithms.							
7.	To familiarize with TCP and UDP header formats.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							K L
1.	Explain the overview of Data communication principles.							K2
2.	Explain the concepts of layered architecture of the OSI model and TCP/IP model, and the concepts of switching and multiplexing techniques.							K2
3.	Analyze flow control, error control and access control issues.							K3
4.	Analyze the operation of different network devices, routing, congestion control algorithms, IP protocol and IP addressing.							K3
5.	Analyze the performance of transport layer and application layer protocols							K3
SYLLABUS								
UNIT-I (12 Hrs)	Data Communication Fundamentals: Data Representation, Data Transmission, Modes of Data Transmission, Signal Encoding; Modems and Data Multiplexers: Introduction to Digital Modulation Methods, Multilevel Modulation, Differential PSK, Standard Modems. Uses of Computer Networks, Line Configuration, Topology, Transmission mode, Categories of Networks-LAN, MAN, WAN.							
UNIT-II (12 Hrs)	Layered architecture: Protocol Hierarchies, Design issues of layers, Connection Oriented and Connectionless services; Reference Models- The OSI Reference Model, The TCP/IP Reference Model, The B-ISDN ATM Reference Model. physical layer: Transmission Media, Multiplexing-Frequency Division Multiplexing, Time Division Multiplexing, Switching- Circuit Switching, packet switching techniques.							
UNIT-III (9 Hrs)	Data link layer Design issues of Data link layer, Error Detection and Correction techniques, Elementary Data link protocols, Sliding window protocols, HDLC, Medium access sub layer-The Channel allocation problem, Multiple Access Protocols-ALOHA, Carrier Sense Multiple Access protocols; IEEE standard for 802 LANs-802.3,Wireless LANs- Wi-Fi 802.11 architecture, MAC protocol.							

UNIT-IV (8 Hrs)	Network layer Virtual circuit and Datagram subnet, Network devices, Routing Algorithms-Distant vector routing algorithm, link state routing algorithm, Congestion Control algorithms-General principles of Congestion Control, Congestion prevention policies. The Leaky bucket algorithm and Token bucket algorithm, The Network Layer in the Internet- The IP Protocol, IP Addresses.
UNIT-V (7 Hrs)	Transport layer The Transport layer Service, Elements of Transport protocols, The Internet Transport Protocols- UDP, TCP. Application layer The Domain Name System, Electronic mail, The World Wide Web.
Text Books:	
1.	Data Communications and Networking by Behrouz A. Forouzan, 2nd edition, Tata McGraw Hill.
2.	Computer Networks — Andrew S Tanenbaum, 3rd Edition, Pearson Education/PHI.
Reference Books :	
1.	An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education
2.	Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson
3	Data Communications and Computer Networks by Prakash.C. Gupta, Prentice- Hall of India Pvt. Ltd
e-Resources:	
1.	https://www.youtube.com/channel/UCr0Ze4SR3MHXAgz1TvRYL7Q
2.	Demonstrations of network configurations, OSI and TCP/IP layered architecture using Cisco packet tracer simulation software.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3107	PC	3	--	--	3	25	75	3 Hrs.
DIGITAL SYSTEM DESIGN USING HDL								
(For ECE)								
Course Objectives:								
1.	Learn the role of HDL in digital system design using VHDL and Verilog programming.							
2.	Understand different levels of abstraction in both VHDL and Verilog HDL.							
3.	Program both combinational and sequential logic circuits in both VHDL and Verilog HDL.							
4.	Understand the operation of state machines and testing of sequential circuits							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Describe and test digital logic circuits in data flow description, structural description, behavioral description and advanced constructs using VHDL.							K3
2.	Describe and test digital logic circuits in data flow description, structural description, behavioral description and advanced constructs using both Verilog.							K3
3.	Design complex sequential state machines that work.							K4
4.	To Apply the overall knowledge of digital circuit design to testing of digital circuits.							K4
SYLLABUS								
UNIT-I (8 Hrs)	Digital Logic Design using VHDL Introduction, designing with VHDL, design entry methods, logic synthesis, entities, architecture, packages and configurations, types of models: dataflow, behavioral, structural, signals vs. variables, generics, data types, concurrent vs. sequential statements, loops and program controls.							
UNIT-II (12Hrs)	Digital Logic Circuit Design Examples using VHDL Combinational Logic circuits building blocks: Multiplexers, Decoders, Encoders, Code converters, Arithmetic comparison circuits, VHDL for combinational circuits, Adders- Half Adder, Full Adder, Ripple-Carry Adder, Carry Look-Ahead Adder, Subtraction, Multiplication. Sequential Logic circuits building blocks: Flip-flops, registers & counters, synchronous sequential circuits, Basic design steps, Mealy State model, Design of FSM using CAD tools, Serial Adder Example, State Minimization, Design of Counter using sequential Circuit approach.							
UNIT-III (8 Hrs)	Digital Logic Design using Verilog HDL Introduction, Verilog Data types and Operators, Binary data manipulation, Combinational and Sequential logic design, Structural Models of Combinational Logic, Logic Simulation, Design Verification and Test Methodology, Propagation Delay, Truth Table models using Verilog.							
UNIT-IV (12 Hrs)	Digital Logic Circuit Design Examples using Verilog HDL Behavioral modeling, Data types, Boolean-Equation-Based behavioral models of combinational logics, Propagation delay and continuous assignments, latches and level-							

	sensitive circuits in Verilog, Cyclic behavioral models of flip-flops and latches and Edge detection, comparison of styles for behavioral model; Behavioral model, Multiplexers, Encoders and Decoders, Counters, Shift Registers, Register files, Dataflow models of a linear feedback shift register.
UNIT-V (8 Hrs)	Testing of Digital Logic Circuit Design Machines with multi cycle operations, ASM and ASMD charts for behavioral modeling. Testing of logic circuits, fault model, complexity of a test set, path-sensitization, circuits with tree structure, random tests, testing of sequential circuits, built in self-test.
Text Books:	
1.	Stephen Brown & Zvonko Vranesic, "Fundamentals of Digital logic design with VHDL", Tata McGraw Hill, 2nd edition.
2.	Michael D. Ciletti, "Advanced digital design with the Verilog HDL", Eastern economy edition, PHI.
Reference Books:	
1.	Stephen Brown & Zvonko Vranesic, "Fundamentals of Digital logic with Verilog design", Tata McGraw Hill, 2nd edition.
2.	Bhaskar, "VHDL Primer", 3rd Edition, PHI Publications.
e-Resources:	
1.	https://nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs72/

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EC3108	PE	3	--	--	3	25	75	3 Hrs.	
SOFT COMPUTING TECHNIQUES									
(For ECE)									
Course Objectives:									
1.	To study the soft computing basics and branches.								
2.	To Understand the basic implementation details on Artificial Neural Networks								
3.	To understand fuzzy logic and it application in ANN.								
4.	To study the Genetic algorithm and its application.								
5.	To study the applications of Soft Computing.								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								KL
1.	Differentiate between Soft Computing and hard computing								K2
2.	Understand and apply Artificial Neural Networks, Fuzzy Logic, and Genetic algorithms for different applications								K3
3.	Understand various applications of soft computing.								K3
SYLLABUS									
UNIT-I (8 Hrs)	Introduction: Approaches to intelligent control, Architecture for intelligent control, Symbolic reasoning system, Rule-based systems, the AI approach, Knowledge representation - Expert systems.								
UNIT-II (12Hrs)	Artificial Neural Networks: Concept of Artificial Neural Networks and its basic mathematical model, McCulloch-Pitts neuron model, simple perceptron, Adaline and Madaline, Feed-forward Multilayer Perceptron, Learning and Training the neural network, Data Processing: Scaling, Fourier transformation, principal-component analysis, and wavelet transformations, Hopfield network, Self-organizing network and Recurrent network, Neural Network based controller.								
UNIT-III (8 Hrs)	Fuzzy Logic System: Introduction to crisp sets and fuzzy sets, basic fuzzy set operation and approximate reasoning, Introduction to fuzzy logic modeling and control, Fuzzification, inferencing and defuzzification, Fuzzy knowledge and rule bases, Fuzzy modeling and control schemes for nonlinear systems, Self-organizing fuzzy logic control, Fuzzy logic control for nonlinear time-delay system.								
UNIT-IV (12 Hrs)	Genetic Algorithm: The basic concept of Genetic algorithm and detail algorithmic steps, Adjustment of free parameters, Solution of typical control problems using genetic algorithm, Concept on some other search techniques like Tabu search and ant-colony search techniques for solving optimization problems.								

UNIT-V (8 Hrs)	Applications: GA application to power system optimization problem, Case studies: Identification and control of linear and nonlinear dynamic systems using MATLAB-Neural Network toolbox, Stability analysis of Neural-Network interconnection systems, Implementation of fuzzy logic controller using MATLAB fuzzy-logic toolbox, Stability analysis of fuzzy control systems.
Text Books:	
1.	Introduction to Artificial Neural Systems - Jacek.M.Zurada, Jaico Publishing House, 1999.
2.	Neural Networks and Fuzzy Systems - Kosko, B., Prentice-Hall of India Pvt. Ltd., 1994.
Reference Books:	
1.	Introduction Neural Networks Using MATLAB 6.0 - S.N. Shivanandam, S. Sumati, S. N. Deepa,1/e, TMH, New Delhi.
2.	Elements of Artificial Neural Networks - KishanMehrotra, Chelkuri K. Mohan, Sanjay Ranka, Penram International.

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EC3109	PC	3	--	--	3	25	75	3 Hrs.	
OPERATING SYSTEMS									
(For ECE)									
1.	Introduce to the internal operation of modern operating systems								
2.	Define, explain, processes and threads, mutual exclusion, CPU scheduling, deadlock, memory management, and file systems								
3.	Understand File Systems in Operating System like UNIX/Linux and Windows								
4.	Understand Input Output Management and use of Device Driver and Secondary Storage (Disk) Mechanism								
5	Analyze Security and Protection Mechanism in Operating System								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								KL
1.	Describe various generations of Operating System and functions of Operating System, System calls								K2
2.	Describe the concept of process, threads and analyze various CPU Scheduling Algorithms and IPC								K2
3.	Illustrate memory management strategies								K3
4.	Illustrate deadlocks, files and Secondary-Storage Structure								K3
5.	Summarize Security and Protection Mechanism in Operating Systems. Understand the Operating System like UNIX/Linux and Windows								K3
SYLLABUS									
UNIT-I (6Hrs)	Operating Systems Overview: Operating system functions, Operating system structure, Operating systems operations, Computing environments, Open-Source Operating Systems. System Structures: Operating System Services, User and Operating-System Interface, systems calls, Types of System Calls, system programs, operating system structure, operating system debugging, System Boot.								
UNIT-II (10Hrs)	Process Concept: Process scheduling, Operations on processes, Inter-process communication, Communication in client server systems. Multithreaded Programming: Multithreading models, Thread libraries, Threading issues. Process Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling, Thread scheduling. Inter-process Communication: Race conditions, Critical Regions, Mutual exclusion with busy waiting, Sleep and wakeup, Semaphores, Mutexes, Monitors, Message passing, Barriers, Classical IPC Problems - Dining philosophers problem, Readers and writers problem.								

UNIT-III (06Hrs)	Memory-Management Strategies: Introduction, Swapping, Contiguous memory allocation, Paging, Segmentation. Virtual Memory Management: Introduction, Demand paging, Copy on-write, Page replacement, Page replacement Algorithms, Frame allocation, Thrashing, Memory-mapped files, Kernel memory allocation.
UNIT-IV (10Hrs)	Deadlocks: Resources, Conditions for resource deadlocks, Deadlock detection and recovery, Deadlock avoidance, Deadlock prevention. File Systems: Files, Directories, File system implementation, management and optimization. Secondary-Storage Structure: Overview of disk structure, and attachment, Disk scheduling, RAID structure, Stable storage implementation.
UNIT-V (08Hrs)	System Protection: Goals of protection, Principles and domain of protection, Access matrix, Access control, Revocation of access rights. System Security: Introduction, Program threats, System and network threats. Case Studies: Linux, Microsoft Windows.
Text Books:	
1.	Silberschatz A, Galvin P B, and Gagne G, Operating System Concepts, 9th edition, Wiley, 2013.
2.	Tanenbaum A S, Modern Operating Systems, 3rd edition, Pearson Education, 2008. (for Interprocess Communication and File systems.)
Reference Books:	
1.	Dhamdhere D M, Operating Systems A Concept Based Approach, 3rd edition, Tata McGraw-Hill, 2012.
2.	Stallings W, Operating Systems -Internals and Design Principles, 6th edition, Pearson Education, 2009
3.	Nutt G, Operating Systems, 3rd edition, Pearson Education, 2004
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105214/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3110	PC	--	--	3	1.5	20	30	3 Hrs.
DIGITAL COMMUNICATION LAB (WITH SIMULATION)								
(For ECE)								
Course Objectives:								
1.	The purpose of this course is to provide the student with a practical perspective of various digital communication modules							
2.	Simplify the practical illustrations of various Digital Modulation and demodulation techniques.							
3.	This lab focuses on the fundamental concepts of Sampling, Pulse modulations, Digital modulation techniques.							
4.	To be familiar with Spread spectrum modulation and demodulation techniques.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Design, implement and verify the theoretical concepts of sampling practically.							K3
2.	Analyze and implement analog to digital converters like PCM, DM.							K4
3.	Comprehend the design, application and practical implementation of various Digital Modulation techniques.							K5
4.	Analyze digital modulation techniques using MATLAB tools.							K4
LIST OF EXPERIMENTS								
HARDWARE								
1	Sampling Theorem – Verification (Hardware and MATLAB Simulation)							
2	Quantization of Signals (MATLAB Simulation)							
3	Pulse Code Modulation (Hardware and MATLAB Simulation)							
4	Companding(MATLAB Simulation)							
5	Differential Pulse Code Modulation. (Hardware and MATLAB Simulation)							
6	Delta Modulation. (Hardware and MATLAB Simulation)							
7	Adaptive Delta Modulation. (MATLAB Simulation))							
8	Amplitude Shift Keying							
9	Frequency Shift Keying							
10	Binary Phase Shift Keying							
11	BPSK Data Transmission over AWGN Channel & BER Performance (Simulation)							
12	Quadrature Phase Shift Keying.							
13	Spread Spectrum Modulation & Demodulation (MATLAB Simulation)							
14	Code Division Multiple Access (MATLAB Simulation)							
15	Generation of PN Sequences (MATLAB Simulation)							
Reference Books:								
1.	Lab Manual							
2.	Principles of Communication Systems, H. Taub and D. L. Schilling, McGraw Hill, 1971.							
3.	Contemporary Communication Systems Using MATLAB,by Gerhard Bauch, John G Proakis, and Masoud Salehi							
4.	Digital Communication Theory,techniques and applications, R.N.Mutagi.							

e-Resources:	
1.	https://www.mathworks.com/matlabcentral/fileexchange/25293-matlab-for-digital-communication
2.	https://www.mathworks.com/products/communications.html

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3111	PC	--	--	3	1.5	20	30	3 Hrs.
LINEAR INTEGRATED CIRCUITS & PULSE CIRCUITS LAB WITH SIMULATION								
(For ECE)								
Course Objectives:								
1.	This laboratory course enables students to get practical experience in design, assembly, and evaluation of Linear integrated circuits & Pulse Circuits. They will use Multisim to test their electronic designs.							
Course outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Design and test different types of Multivibrators							4
2.	Design, construct Schmitt trigger using operational amplifier							4
3.	Observe the operation of clampers and clippers							2
4.	Use Multisim to test their electronic designs							4
LIST OF EXPERIMENTS								
Hardware								
1	Linear Wave Shaping a) Passive RC Differentiator b) Passive RC Integrator							
2	Non-Linear Wave shaping a) Clipping Circuits b) Clamping Circuits							
3	Fixed bias bistable Multivibrator							
4	Schmitt Trigger Using μA 741							
5	Design of Active Filters							
6	Astable Multivibrator using 555 timers							
Simulation								
7	Linear Wave Shaping a) Passive RC Differentiator b) Passive RC Integrator							
8	Non-Linear Wave shaping a) Clipping Circuits b) Clamping Circuits							
9	Fixed bias bistable Multivibrator							
10	Schmitt Trigger Using μA 741							
11	Frequency response of Active Filters							
12	Astable Multivibrator using 555 timers							
Reference Books:								
1.	Linear Integrated Circuits- D.Roy Chowdhury, New Age International(p) Ltd, 2nd Edition,2003							
2.	Pulse and Digital Circuits, A. Anand Kumar, PHI, Second Edition, 2005							
e-Resources:								
1.	<u>Integrated Circuits, MOSFETs, Op-Amps and their Applications</u> https://nptel.ac.in/courses/108/108/108108111/							
2.	<u>Op-Amp Practical Applications: Design, Simulation and Implementation</u> https://nptel.ac.in/courses/108/108/108108114/							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS –I								
Common to All Branches								
Part-A: Verbal and Soft Skills-I								
Course Objectives:								
1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.							
2.	To familiarize the learner with high frequency words as they would be used in their professional career.							
3.	To inculcate logical thinking in order to frame and use data as per the requirement							
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.							
5.	To familiarize students with soft skills and how it influences their professional growth.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.							K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.							K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.							K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.							K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.							K3
SYLLABUS								
UNIT-I	Grammar: (VA) Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns- number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.							
UNIT-II	Vocabulary: (VA) Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard							

	to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.
UNIT-III	Reasoning: (VA) Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).
UNIT-IV	Usage: (VA) Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	Soft Skills: Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners,,s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal,,s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.

Part-B: Quantitative Aptitude-I		
Course Objectives:		
1.	To familiarize students with basic problems on numbers and ratios problems.	
2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.	
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.	
4.	To inculcate logical thinking by exposing the students to reasoning related questions.	
5.	To inculcate logical thinking by exposing the students to reasoning related questions.	
Course Outcomes: At the end of this course, the students will be able to		
S.No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	

UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3102	MC	--	--	3	--	--	--	--
BASIC CODING								
(Common to ECE and EEE)								
Course Objectives:								
1.	To develop programming skills among the students							
2.	To familiarize the student with Control Structures, Loop Structures							
3.	To familiarize the student with Basic searching and sorting Methods							
4.	To familiarize the student with Functions, Recursions and Storage Classes							
5.	To familiarize the student with Structures and Unions							
6.	. To familiarize the student with Operating System concepts.							
7.	To familiarize the student with Networking concepts.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Know about Control Structures, Loop Structures and branching in programming							K3
2	Know about various searching and sorting methods.							K3
3.	Know about Functions, Recursions and Storage Classes							K3
4.	Know about Structures and Unions.							K3
5.	Know different Operating System concepts. 6.							K3
6.	Differentiate OSI Model Vs. TCP/IP suite.							K3
SYLLABUS								
UNIT-I	Review of Programming constructs Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.							
UNIT-II	Introduction to Linear Data, strings and pointers Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.							
UNIT-III	Functions, Recursions and Storage Classes Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions. Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime- memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.							

UNIT-IV	User-defined data types, Pre-processor Directives and standard storage Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming. Practice: Structure padding, user-defined data storage and retrieval programs
UNIT-V	Operating system principles and Database concepts: Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer.
Reference Books:	
1.	Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2.	The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd.
3.	Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific
4.	Object Oriented Programming in C++: N. Barkakati, PHI.
5.	Object Oriented Programming through C++ by RobatLaphore.
6.	https://www.geeksforgeeks.org/ .
7.	https://www.tutorialspoint.com/

ELECTRONICS & COMMUNICATION ENGINEERING
(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

II-SEMESTER

(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 EC 3201	Microprocessors & Microcontrollers	PC	3	3	--	--	25	75	100
B19 EC 3202	Digital Signal Processing	PC	3	3	--	--	25	75	100
B19 EC 3203	VLSI Design	PC	3	3	--	--	25	75	100
# PE-II	Professional Elective-II	PE	3	3	--	--	25	75	100
# OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 EC 3208	Microprocessors & Microcontrollers Lab	PC	1.5	--	--	3	20	30	50
B19 EC 3209	DSP - Lab	PC	1.5	--	--	3	20	30	50
B19 EC 3210	VLSI - Lab	PC	1.5	--	--	3	20	30	50
B19 EC 3211	Mini Project/ Internship	PR	1.5	--	--	3	--	50	50
B19 MC 3201	Employability Skills-II	MC	--	3	--	--	--	--	--
B19 MC 3203	Advanced Coding	MC	--	--	--	3	--	--	--
TOTAL			21	18	--	15	185	515	700

	Course Code	Course
#PE-II	B19 EC 3204	Information Theory and Coding
	B19 EC 3205	Digital IC Design
	B19 EC 3206	Digital Signal Processors and Architectures
	B19 EC 3207	Data Base Management Systems
#OE-I	Student has to study one Open Elective offered by CE or CSE or EEE or IT or ME or EM &H from the list enclosed.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3201	PC	3	--	--	3	25	75	3 Hrs.
MICROPROCESSORS AND MICROCONTROLLERS								
(For ECE)								
Course Objectives:								
1.	To understand the architecture of 8085 Microprocessor							
2.	To be familiar with 8085 assembly language programming							
3.	To understand the architecture of 8086/8088 Microprocessor							
4.	To be familiar with 8086 assembly language programming.							
5.	To understand the architecture of 8051 Microcontroller and be familiar with assembly language programming.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1	Illustrate architecture of the 8085 microprocessors.							K3
2	Develop 8085 Microprocessor Assembly Language Programs.							K3
3.	Illustrate architecture of the 8086 microprocessors.							K2
4.	Develop 8086 Microprocessor Assembly Language Programs.							K4
5.	Describe the functional block of 8051 microcontroller and Implement Assembly Language Programs							K3
SYLLABUS								
UNIT-I (10 Hrs)		8085 Architecture: Introduction to Intel 8 bit Microprocessors, Bus structure of 8085, internal architecture and functional description of INTEL 8085 Microprocessor pin out & signals, flag register, Fetch cycle, memory Read /Write and I/O Read /Write Cycles with Timing Diagrams, Stack memory organization, Interrupt structure of 8085, Vectored, non-vectored, maskable and non-maskable interrupts, pending interrupts, execution of SIM and RIM instructions.						
UNIT-II (9 Hrs)		8085 Programming: Introduction to 8085 Assembly Language Programming, Programming model of 8085 and function of each register, Addressing modes of 8085 with examples, I/O addressing, IN & OUT instructions with timing diagrams, Stack memory operation using PUSH and POP instructions, Classification of 8085 instructions based on length and operation with examples, Sample Programs using Data Transfer Instructions, Arithmetic Instructions,BCD addition using DAA, Delay programs using Subroutines, Execution of CALL and RET instructions with Timing diagrams.						
UNIT-III (10 Hrs)		8086/8088 Architecture:Introduction to INTEL 16-bit Microprocessors, Internal Architecture and Functional description of Intel 8086/8088 microprocessors, and their comparisons. Memory segmentation and physical memory address generation, pipeline architecture and instruction queue. Register organisation, Status flags and machine control flags of 8086, pin out and signals in detail, Memory read /write and I/O read/Write Bus cycles with timing diagrams, 8086 memory Banks.						

UNIT-IV (9 Hrs)	8086 Programming: Introduction to 8086 Assembly language programming, programmable register array of 8086 and function of each register, Data addressing modes of 8086 with examples, fixed and variable I/O addressing Using IN & OUT instructions. Stack memory operation Using PUSH and POP instructions, Sample 8086 assembly language programs using data transfer instructions, Arithmetic instructions, DAA and DAS instructions.
UNIT-V (10 Hrs)	Introduction to Microcontrollers (8051): Microprocessors & Microcontrollers Comparison, Overview of 8051 Microcontroller, Internal Block Diagram of 8051, Memory Organization, Internal RAM Memory Structure, on chip registers of 8051, Program Status Word (PSW) Flag Register, Addressing Modes and Sample 8051 Assembly Language Programs using Data Transfer, arithmetic and logical Instructions.
Text Books:	
1.	Architecture Programming and Applications. Ramesh S.Goankar.New Age International Pvt.Ltd.,(3rd Edition)'
2.	Microprocessors and interfacing ,Douglas V.Hall, Tata McGraw-Hill Revised 2nd Edition.
3.	The 8051 Microcontroller and Embedded Systems using assembly and C- Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D.McKinlay;PHI,2013/ Pearson, 2013
Reference Books:	
1.	Microprocessors: The 8086/8088, 80186/80286, 80386/80486 and the Pentium Family .Nilesh B.Bahadure, Phi Learning Pvt.Ltd.,2010
2.	Kenneth J Ayala, The 8051 Microcontroller Architecture Programming and Application, third Edition, Penram International Publishers
e-Resources:	
1.	https://nptel.ac.in/courses/108/105/108105102/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3202	PC	3	--	--	3	25	75	3 Hrs.
DIGITAL SIGNAL PROCESSING								
(For ECE)								
Course Objectives:								
1.	This course introduces students to the fundamental principles of Digital Signal Processing and develops essential analysis and design tools required for signal processing systems & implementations. Also, this subject is an introduction to the graduate-level courses in a broad range of disciplines spanning communications, speech processing & image processing. The topics include SS basics, sampling theorem, Z-transform, analysis of Discrete-time Linear Time-Invariant Systems, Realization structures, Frequency domain representation of signals and systems, DTFT, DFS, Discrete Fourier Transform (DFT), linear/circular convolutions, Fast Fourier Transform (FFT) algorithms, FIR & IIR digital filter design, Multi-rate DSP and a few DSP applications							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Describe the DSP fundamental theory and components, Develop an understanding of DSP advantages, limitations and fundamental tradeoffs. Carry-out LTI system analysis using convolution & Z-transform							K3
2.	Carryout data analysis & spectrum analysis using FFT							K4
3.	Design of IIR digital filters to meet specifications							K4
4.	Design of FIR digital filters to meet specifications							K4
5.	Knows multi-rate signal processing aspects & DSP applications							K3
SYLLABUS								
UNIT-I (10 Hrs)	Discrete-Time Signals and Systems: (Oppenheim & Proakis) Introduction to Digital Signal Processing, Basic elements of a DSP system, Advantages of Digital SP over Analogy SP, Discrete-time signals and systems, DT-LTI systems described by Linear constant-coefficient difference equations, Properties & Analysis of DT-LTI systems, Discrete linear convolution, Frequency domain representation of DT Signals and Systems, DTFT, Review of the Z-transform, Properties, Inverse Z-transform, Analysis of DT LTI systems in Z-Domain, System function, One-sided Z-transform, Solution of difference equations, Structures and Realization of Digital Filters, Direct-I, II, series and parallel forms.							
UNIT-II (10 Hrs)	Discrete Fourier Transform (DFT) and Fast Fourier Transform Algorithms (FFT): (Oppenheim & Proakis) Frequency analysis of discrete time signals, DFS, Properties of DFS, Sampling of DTFT, DFT, Properties of DFT, Circular and linear convolution of sequences using DFT, Efficient computation of DFT, Radix-2 Decimation-in-Time(DIT) & Decimation-in-Frequency(DIF) FFT Algorithms, Inverse FFT							
UNIT-III (8 Hrs)	Design of IIR Digital Filters: (Oppenheim & Proakis) General considerations in Filter design, Analog filter approximations- Butterworth and Chebyshev, Frequency response specifications; Design of IIR digital filters from analog filters, Bilinear Transformation Method, Impulse Invariance Technique, and Low-pass filter Design examples.							

UNIT-IV (8 Hrs)	Design of FIR Digital Filters: (Oppenheim & Proakis) Characteristics of FIR Digital Filters, Design of Linear Phase FIR digital Filters using Windows, Effect of Window selection & filter length on filter frequency response, Design examples, Comparison of IIR and FIR Filters.
UNIT-V (10 Hrs)	DSP Applications and Fundamentals of Multirate Digital Signal Processing: (SK Mitra) Overview of DSP applications, Spectral analysis of sinusoidal signals using FFT, Subband coding of speech signals, Signal compression, Finite precision arithmetic effects. Introduction to Multirate DSP, Basic sampling rate alteration devices: up sampler, down sampler, Time and Frequency domain characterization of up/down samplers, Interpolator and decimator. Interactive programming-based examples.
Text Books:	
1.	Alan V. Oppenheim, Ronald W. Schaffer, —Digital Signal Processing— PHI Ed., 2006
2.	John G. Proakis, D.G. Manolakis, —Digital Signal Processing: Principles, Algorithms and Applications—, 3rd Ed., PHI, 1996.
Reference Books:	
1.	DSP - NPTEL Video course by Prof. S.C. Dutta Roy, IIT Delhi. https://nptel.ac.in/courses/117/102/117102060/
2.	https://ocw.mit.edu/resources/res-6-008-digital-signal-processing-spring-2011/ .
3.	https://nptel.ac.in/courses/108/105/108105055/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3203	PC	3	--	--	3	25	75	3 Hrs.
VLSI DESIGN								
(For ECE)								
Course Objectives:								
1.	To introduce various fabrication steps of MOS transistors and their electrical properties.							
2.	To implement the stick diagrams and layouts using CMOS/Bi-CMOS design rules.							
3.	To explain MOS technology interconnection as circuits, scaling models, static and dynamic designs.							
4.	To introduce the concepts of FPGA and testing methods of digital circuits.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Analyze the Electrical properties and Fabrication processes of MOS circuits.							K3
2.	Design the layouts of various MOS circuits by applying the concept of design rules.							K4
3.	Interpret the basic MOS circuit concepts, static and dynamic CMOS logic designs and the impact of scaling on MOS circuits.							K3
4.	Analyze various testing methods of digital circuits and the basic concepts of FPGA.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Introduction to IC Technology, Fabrication process: NMOS, PMOS and CMOS. Ids versus Vds Relationships, Aspects of MOS transistor Threshold Voltage, MOS transistor Transconductance, Output Conductance and Figure of Merit. NMOS Inverter, Pull-up to Pull down Ratio for NMOS inverter driven by another NMOS Inverter, and through one or more pass transistors, Alternative forms of pull-up, The CMOS Inverter, Latch-up in CMOS circuits, Comparison between CMOS and Bi-CMOS technology.							
UNIT-II (5 Hrs)	MOS and Bi-CMOS Circuit Design Processes: MOS Layers, Stick Diagrams, Design Rules and Layout, General observations on the Design rules, 2µm Double Metal, Double Poly, CMOS/BiCMOS rules, 1.2µm Double Metal, Double Poly CMOS rules, Layout Diagrams of NAND and NOR gates and CMOS inverter.							
UNIT-III (8 Hrs)	Basic Circuit Concepts: Sheet Resistance, Sheet Resistance concept applied to MOS transistors and Inverters, Area Capacitance of Layers, Standard unit of capacitance, The Delay Unit, Inverter Delays, Propagation Delays, Wiring Capacitances, Choice of layers. Scaling of MOS Circuits: Scaling models, Scaling factors for device parameters, Limitations of Scaling on substrate doping, Miniaturization, Interconnect and contact Resistance, Sub threshold currents and current density.							
UNIT-IV (10 Hrs)	CMOS Combinational and Sequential logic circuit design: Static CMOS Design: Complementary CMOS and its static properties, Ratioed logic, Pass Transistor logic- Design of logic gates. Dynamic CMOS Design: Basic principles, speed and power dissipation of dynamic logic, Issues in dynamic logic- charge leakage, charge sharing, Static latches and registers- Latches versus registers, bistability principle, SR- Flip flops, Multiplexer based latch, Master-slave-edge triggered register.							

UNIT-V (8 Hrs)	FPGA Design: FPGA design flow, Basic FPGA architecture, FPGA Technologies, Introduction to FPGA Families. Test and Testability: Design for Testability-Path sensitization, Scan Design Techniques- Scan path, Level sensitive scan design (LSSD), Boundary scan test (BST) and Built-In-Self Test.
Text Books:	
1.	Essentials of VLSI Circuits and Systems By Kamran Eshraghian, Douglas and A. Pucknell and SholehEshraghian, Prentice-Hall of India Private Limited,2005 Edition.
2.	Digital Integrated Circuits, Jan M. Rabaey, AnanthaChandrakasan and Borivoje Nikolic,2nd edition, 2016
Reference Books:	
1.	FPGA Based System Design- Wayne Wolf, Pearson Education, 2004, Technology and Engineering
2.	CMOS Digital Integrated Circuits Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, Tata McGraw Hill Education,2003.
Web Links:	
1.	https://www.engineersgarage.com/vlsi-technology-an-overview/
2.	https://www.tutorialspoint.com/vlsi_design/vlsi_design_digital_system.htm
3.	https://www.powershow.com/viewfl/e5a26-ZDc1Z/Lecture 4 Design Rules Layout and Stick Diagram powerpoint ppt presentation

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3204	PE	3	--	--	3	25	75	3 Hrs.
INFORMATION THEORY AND CODING								
(For ECE)								
Course Objectives:								
1.	To introduce students to the basic concepts of information theory & and applications of coding for information compression & channel protection, entropy, source coding and the fundamental limits of data compression							
2.	To study mutual entropy & channel capacity to understand the fundamental limits of information transmission, channel coding & decoding techniques for error control and means to achieve it.							
3.	To give a foundation for studying modern wireless communications & information security.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Appreciate the mathematical concept of information (uncertainty) via probability, compute the entropy of a source &Outline the need of source coding & variable length codes.							K3
2.	Device source codes using Shannon-Fano & Huffman algorithms, calculate the efficiency of a code							K3
3.	Compute mutual entropy of a channel, understand the concept of channel capacity, State Shannon’s noisy channel coding theorem which creates the field of channel coding, compute channel capacity of BSC & AWGN channels, define characteristics of an ideal communication system.							K4
4.	Realize the need & benefits of channel coding, Outline Linear block codes structure, theory &use syndrome technique for decoding for linear block codes, Study cyclic codes (BCH, RS and CRC) structure, theory, implementation & decoding of cyclic codes, differentiate source coding and channel coding &learn applications of coding							K4
5.	Study Convolutional codes representation, generation & decoding of convolutional codes using Viterbi algorithm, get acquainted with concatenated codes to increase coding gain & Trellis Coded Modulation (TCM), Know modern codes & pursue modern wireless communications & information security courses.							K2
SYLLABUS								
UNIT-I (8 Hrs)	PART-I: INFORMATION THEORY (Carlson &Haykin books) Information Theory and Source Coding:Discrete memory less sources, Information measure, Entropy and Information rate, Shannon’s source coding theorem, Coding for a discrete memory less source, Shannon-Fano & Huffman algorithms, Predictive coding for sources with memory.							
UNIT-II (8 Hrs)	Information Transmission:Mutual information, Mutual entropy, Discrete channel capacity, Shannon’s channel coding theorem, Coding for the Binary Symmetric Channel, Continuous channels: Continuous information, Entropy, Entropy maximization, AWGN channel capacity, Ideal communication system.							

UNIT-III (8 Hrs)	PART-II: ERROR CONTROL CODING (Haykin& K. Deergha Rao books) Channel Coding: Block Codes & Cyclic Codes: Rationale for coding, Types of codes, Discrete memory less channels, Linear block codes, Syndrome decoding, Cyclic codes, Properties of BCH, RS and CRC codes.
UNIT-IV (8 Hrs)	Convolutional Codes & Trellis codes: Representation & generation, Decoding Convolutional codes, Exhaustive search method, Maximum Likelihood decoding of Convolutional codes, Viterbi Algorithm, Sequential decoding, Trellis codes, Burst error correction, Interleaving, Automatic Repeat Request (ARQ) schemes.
UNIT-V (8 Hrs)	Modern Codes: Applications of coding, Concatenated coding, Turbo codes, Non-recursive and Recursive Systematic Convolutional(RSC) Encoders, Turbo Encoder, Low Density Parity Check(LDPC) Codes, Properties, Parity Check Matrix H, Tanner Graphs, MIMO System, Space–Time-Coded MIMO System, Space–Time Block Codes (STBC), Alamouti 2-transmit Code (2-Transmit, 1- Recieve)
Text Books:	
1.	Communication Systems, 3/e, by A.B. Carlson, Mcgraw Hill Publishers (Part-I)
2.	Digital Communications by Simon Haykin, John Wiley & Sons (Part-I & II)
Reference Books:	
1.	Channel Coding Techniques for Wireless Communications, by K.Deergha Rao, Springer publications (part-II)
2.	Information Theory, Coding and Cryptography – Ranjan Bose, 2 nd Ed, 2009, TMH.
Web Links:	
1	https://nptel.ac.in/courses/117/101/117101053

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3205	PC	3	--	--	3	25	75	3 Hrs.
DIGITAL IC DESIGN								
(For ECE)								
Course Objectives:								
1.	To understand the MOS Design.							
2.	To design Combinational and Sequential MOS Logic Circuits.							
3.	To provide insights of the Digital Integrated Circuits for different Applications.							
4.	To introduce the concepts of Semiconductor Memories, Flash Memory, RAM array organization.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Interpret the concepts of MOS Design.							K3
2.	Analyze different Combinational and Sequential MOS Circuits.							K3
3.	Design and develop the Digital Integrated Circuits for different Applications.							K4
4.	Analyze the Concepts of Semiconductor Memories, Flash Memory, RAM array organization.							K3
SYLLABUS								
UNIT-I (12 Hrs)	MOS Design: Pseudo NMOS Logic – Inverter, Inverter threshold voltage, output high voltage, Output Low voltage, gain at gate threshold voltage, Transient response, Rise time, Fall time, Pseudo NMOS logic gates, Transistor equivalency, CMOS Inverter logic.							
UNIT-II (8 Hrs)	Combinational MOS Logic Circuits: MOS logic circuits with NMOS loads, Primitive CMOS logic gates – NOR & NAND gate, Complex Logic circuits design – Realizing Boolean expressions using NMOS gates and CMOS gates, AOI and OAI gates, CMOS full adder, CMOS transmission gates, Designing with Transmission gates.							
UNIT-III (8 Hrs)	Sequential MOS Logic Circuits: Behavior of bistable elements, SR Latch, Clocked latch and flip flop circuits, CMOS D latch and edge triggered flip- flop.							
UNIT-IV (8 Hrs)	Dynamic Logic Circuits: Basic principle, Voltage Bootstrapping, Synchronous dynamic pass transistor circuits, Dynamic CMOS transmission gate logic, High performance Dynamic CMOS circuits							
UNIT-V (8 Hrs)	Semiconductor Memories: Memory Types, RAM array organization, DRAM – Types, Operation, Leakage currents in DRAM cell and refresh operation, SRAM operation Leakage currents in SRAM cells, Flash Memory- NOR and NAND flash.							
Text Books:								
1.	Digital Integrated Circuits – A Design Perspective, Jan M. Rabaey, AnanthaChandrakasan, BorivojeNikolic, 2nd Ed., PHI.							
2.	Digital Integrated Circuit Design – Ken Martin, Oxford University Press, 2011.							

Reference Books:	
1.	CMOS Digital Integrated Circuits Analysis and Design – Sung-Mo Kang, Yusuf Leblebici, TMH, 3rd Ed., 2011.
2.	CMOS VLSI Design – Neil H.E Weste, David harris, Ayan Banerjee 3rd Edition, Pearson
e-Resources:	
1.	<u>NPTEL :: Electrical Engineering - NOC:Digital IC Design</u>
2.	<u>CMOS Digital VLSI Design - YouTube</u>

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3206	PC	3	--	--	3	25	75	3 Hrs.
DIGITAL SIGNAL PROCESSORS AND ARCHITECTURES								
(For ECE)								
Course Objectives:								
1.	To study the basic DFT, FFT and rate conversion algorithms.							
2.	To study the number format, dynamic range and sources of errors in DSP systems.							
3.	To learn about TMS programmable DSPs and their programming capabilities.							
4.	To understand the interfacing and memory device for TMS processors.							
5.	To study the different applications on DSP Processors.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							KL
1.	Apply DFT and FFT algorithms for DSP application							K3
2.	Apply the number format, dynamic range and various sources of errors in DSP system.							K3
3.	Understand the limitations and capabilities of TMS Processor.							K3
4.	Enhance the applications of DSP Processors by interface with different devices.							K4
5.	Implement different applications on DSP Processors.							K4
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to Digital Signal Processing: Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences. Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation.							
	Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.							
UNIT-II (8 Hrs)	Architectures for Programmable DSP Devices: Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation UNIT, Programmability and Program Execution, Speed Issues, Features for External interfacing.							
UNIT-III (8 Hrs)	Programmable Digital Signal Processors: Commercial Digital signal-processing Devices, Data Addressing modes of TMS320C54XX DSPs, Data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX Processors, Program Control, MS320C54XX instructions and Programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, Pipeline operation of TMS320C54XX Processors.							

UNIT-IV (6 Hrs)	Interfacing Memory and I/O Peripherals to Programmable DSP Devices: Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).
UNIT-V (10 Hrs)	Applications of Programmable DSP Devices: Introduction, A DSP system, DSP Based Biotelemetry Receiver, Complete Speech Processing System, Image Processing System,
Text Books:	
1.	Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2.	A Practical Approach to Digital Signal Processing - K Padmanabhan, R. Vijayarajeswaran, Ananthi. S, New Age International, 2006/2009
3.	EmbeddedSignalProcessingwiththeMicroSignalArchitecturePublisher: Woon-SengGan, Sen M. Kuo, Wiley-IEEE Press, 2007
Reference Books:	
1.	Digital Signal Processors, Architecture, Programming and Applications – B. Venkataramani and M. Bhaskar, 2002, TMH.
2.	DSP Processor Fundamentals, Architectures & Features – Lapsley et al. 2000, S. Chand & Co.
3.	Digital Signal Processing Applications Using the ADSP-2100 Family by The Applications Engineering Staff of Analog Devices, DSP Division, Edited by Amy Mar, PHI
4.	Digital Signal Processing Applications Using the ADSP-2100 Family by The Applications Engineering Staff of Analog Devices, DSP Division, Edited by Amy Mar, PHI
5.	<i>The Scientist and Engineer's Guide to Digital Signal Processing</i> by Steven W. Smith, Ph.D., California Technical Publishing, ISBN 0-9660176-3-3, 1997

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3207	PE	3	--	--	3	25	75	3 Hrs.
DATA BASE MANAGEMENT SYSTEMS								
(For ECE)								
Course Objectives:								
1.	To introduce about database management systems							
2.	To give a good formal foundation on the relational model							
3.	To introduce the concepts of basic SQL as a universal Database language							
4.	To demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design and normalization							
5.	To provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques							
6.	To explain Transaction management techniques							
Course Outcomes: At the end of the course Students will be able								
S.No	Outcome							KL
1.	Describe fundamental concepts a relational database							K2
2.	Create, maintain and manipulate a relational database using SQL							K3
3.	Apply Conceptual and Logical database design							K3
4.	Apply normalization for database design							K3
5.	Illustrate Storage management and Transaction management techniques.							K2
SYLLABUS								
UNIT-I (8Hrs)	Introduction: Database system, Characteristics (Database Vs File System), Database Users (Actors on Scene, Workers behind the scene), Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.							
UNIT-II (10 Hrs)	Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update), basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String conversion).							
UNIT-III (10 Hrs)	Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, Generalization/specialization, Aggregation, SQL: Creating tables with relationships, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, views(updatable and non-updatable), relational set operations.							

UNIT-IV (10 Hrs)	Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, Closure of functional dependency and attribute closure, Normal forms based on functional dependency (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form (BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF), Fifth Normal Form (5NF).
UNIT-V (10 Hrs)	Transaction Concept: Transaction State, Implementation of Atomicity and Durability, Schedules, Serializability, Recoverability, Implementation of Isolation levels, 2Pl and Time stamp ordering protocols, Failure Classification, Recovery and Atomicity, ARIES Recovery algorithm. Indexing Techniques: Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing, B+ Trees: Searching, Insertion, Deletion
Text Books:	
1.	Database System Concepts by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, 7th Edition, McGraw-Hill Education, 2019.
2.	Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition., McGraw-Hill Education (India), 2014.
Reference Books:	
1.	Database Principles: Fundamentals of Design, Implementation, and Management by Steven Morris, Keeley Crockett, Carlos Coronel, Craig Blewett, Cengage, 2020.
2.	Fundamentals of Database Systems by RamezElmasri, Shamkant B. Navathe, 7th Edition, Pearson Education India, 2015.
3.	Introduction to Database Systems by C J Date, 8th Edition, Pearson Education, 2009.
e-Resources:	
1.	https://nptel.ac.in/courses/106/105/106105175/
2.	https://www.geeksforgeeks.org/introduction-to-nosql/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3208	PC	--	--	3	1.5	20	30	3 Hrs.
MICROPROCESSORS & MICROCONTROLLERS LAB								
(For ECE)								
Course Objectives:								
1	To Introduce ALP concepts, features and Coding.							
2	Get familiarized with 8085, 8086 and 8051 instructions.							
3	Developing Assembly Language Programs using the instruction sets of microprocessors (8085,8086) and 8051 microcontrollers.							
4	The accompanying lab is designed to provide practical hands-on experience with Microprocessor hardware kits and software applications							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge level
1.	Develop 8085 assembly language programs on data transfer, arithmetic and logical operations							K3
2.	Develop 8086 assembly language programs using data transfer, arithmetic and logical instructions							K3
3	Develop 8051 assembly language programs on data transfer, arithmetic and logical operations							K3
LIST OF EXPERIMENTS								
Experiments Based On 8085								
1	A). Assume that a byte of data is stored at memory location 'X'. Write an ALP which tests bit 5 Of this data. Write 'FF' in the location 'X+1' if the bit 5 is '1' or Write '00' if bit 5 is '0'. B). Check the zero condition of this number and write '00' at location 'Y' if it is zero or Write 'FF' at 'Y' if it is non-zero. C). For data value in the location 'X' compute the number of logic 1's and store the result in the location 'Y+1'							
2	A). Write an ALP to swap the contents of location 'X' and 'X+1' using BC & HL Register pairs. B). Write an ALP to transfer a block of data from one memory location to another memory location							
3	A). Two 16 bit numbers are stored at memory locations 'X' and 'X+2'.Write an ALP to add these numbers without using DAD instruction and store the result from location Y onwards B). Modify above program to add two 16 bit numbers using DAD instruction.							
4	Two 8 bit numbers 34H and 43H are stored in locations 'X' and 'X+1' compute the product of these two numbers using. A). Repetitive addition method B). Shift and add method							
5	The number of the bytes of a block of data is in location 'X' and data starts from location 'X+1' onwards. By defining a stack pointer, write an ALP to arrange this sequence of data in reverse order. Keep the reverse sequence from 'Y' onwards.							

6	The number of bytes of a block of data is location 'X' and data starts from location 'X+1' onwards. Arrange this block of data in ascending order by using bubble sorting technique
7	Using 8279 write an ALP to generate the message of 4 characters. Activate the LED's individually and make the display ON & OFF for every 0.5 seconds.
Experiments Based On 8086	
1	Write an 8086 ALP to perform addition operation on two-32-bit numbers
2	Write an 8086 ALP to perform Subtraction operation on two-32-bit numbers
3	Write an 8086 ALP to perform Multiplication of two 16-bit numbers
4	Write an 8086 ALP to perform 16-bit Division operation
5	Write an 8086 program to add two four-digit BCD numbers
6	Write an 8086 program to convert 8-bit BCD number into Binary
Experiments Based Microcontroller (8051)	
1	Write an 8051 Assembly Language programs on Data Transfer Instructions
2	Write an 8051 Assembly Language programs on Arithmetic Instructions
3	Write an 8051 Assembly Language programs on logical Instructions
4	Write an 8051 Assembly Language program to find the Average of n-numbers.
5	Write an 8051 Assembly Language program to sum of first 10 Natural Numbers.
Reference Books:	
1.	Lab Manual
e-Resources:	
1.	http://vlabs.iitb.ac.in/vlabs-dev/labs_local/microprocessor/labs/explist.php
2.	http://vlabs.iitb.ac.in/vlabs-dev/labs/8051-Microcontroller-Lab/labs/index.php
3.	https://www.youtube.com/watch?v=HXYhBCpDoVc
4.	https://www.youtube.com/watch?v=xOCjCA3kNgc

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	OE	3	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE-I								
(For ECE)								
Student has to study one Open Elective offered by CE or CSE or EEE or IT or ME or EM&H from the list enclosed.								

Offered by	Course Code	Course Name	Offered to
CIVIL ENGINEERING	B19CEOE01	Disaster Management	CSE,ECE,EEE, IT &ME
	B19CEOE02	Remote Sensing &GIS	
COMPUTER SCIENCE & ENGINEERING	B19CSOE01	Data Structures	CE,ECE,EEE, &ME
	B19CSOE02	Java Programming	
	B19CSOE03	Web Technologies	
ELECTRICAL & ELECTRONICS ENGINEERING	B19EEOE01	Introduction To Electrical Systems	CE,CSE,ECE, IT &ME
	B19EEOE02	Electrical Estimation and Costing	
INFORMATION TECHNOLOGY	B19ITOE01	Data Structures & Algorithms	CE,ECE,EEE, &ME
	B19ITOE02	Database Design & Development	
	B19ITOE03	Java Programming	
MECHNAICAL ENGINEERING	B19MEOE01	Operation Research	CE,CSE,ECE, EEE & IT
	B19MEOE02	Operations Management	
	B19MEOE03	Total Quality Management	
ENGINEERING MATHEMATICS & HUMANITIES	B19BSOE01	computational statistics with R	ALL BRANCHES
	B19BSOE02	Fuzz Sets and Fuzzy Logic	

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EC3209	PC	--	--	3	1.5	20	30	3 Hrs.	
DSP LAB									
(For ECE)									
Course Objectives: Students should learn									
1.	To implement Convolution and Transform techniques on a given sequence								
2.	To analyze frequency response of Discrete Time Signals and Systems								
3.	To design different Digital filters								
4.	To implement basic Image Processing techniques								
Course Outcomes: At the end of the course, students will be able to									
S.No	Outcome								KL
1.	Make use of MATLAB simulation tool for performing various operations on discrete signals.								K6
2.	Make use of MATLAB simulation tool to verify different DSP algorithms.								K6
3.	Make use of MATLAB simulation tool to perform various operations on an Image								K6
LIST OF EXPERIMENTS									
1	Verification of Sampling Theorem								
2	Generation of Discrete Time (DT) Sequences and Signals								
3	Sampling & Quantization of Speech & Image Signals								
4	Linear Convolution & Circular Convolution								
5	Frequency Domain Analysis of DT Signals & Systems using DTFT								
6	LTI System Simulation								
7	Design and Verification of IIR Digital Filters								
8	Design and Verification of FIR Digital Filters using Windows								
9	Spectrum Analysis using Fast Fourier Transform (FFT)								
10	LTI Filtering of Signals								
11	Multirate Up sampling & Down sampling								
12	Design and Simulation of Interpolation & Decimation Filters								
13	Generation & Detection of DTMF Signals								
14	Image intensity transformations								
15	Histogram processing								
16	Filtering images in spatial and frequency domain								
17	Computation of N-point DFT of a given sequence using DSP processor								
18	Computation of FFT of a given sequence using DSP processor								
Reference Books:									
1.	DSP using MATLAB by John G. Proakis.								
2.	Digital Image Processing by Gonzalez and Woods								
3.	Sanjit K. Mitra, "Digital Signal Processing: A Computer Based Approach", Tata McGraw Hill.								
4.	Lab manual								
e-Resources:									
1.	https://www.mathworks.com/academia/books/digital-image-processing-gonzalez.html								
2.	http://vlabs.iitkgp.ernet.in/dsp/index.html								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EC3210	PC	--	--	3	1.5	20	30	3 Hrs.
VLSI LAB								
Course Objectives:								
1.	To learn Verilog /VHDL Source coding, perform simulation and analyze the results using necessary Synthesizer.							
2.	To provide knowledge in designing the schematic diagrams and layouts of various combinational and sequential circuits using CMOS 130nm Technology with necessary EDA tools (Mentor Graphics/Cadence Tools).							
Course Outcomes: At the end of the course, students will be able to								
S.No	Outcome							KL
1.	Analyze and program synthesizable codes in Verilog/VHDL.							K4
2.	Design schematics and layouts using CMOS logic and verify their functionality including parasitic using Cadence/Mentor Graphics CAD tools.							K4
LIST OF EXPERIMENTS								
PART A (Any Seven Experiments)								
Note: Develop Verilog /VHDL Source code, perform simulation using relevant simulator and analyze the obtained simulation results using necessary Synthesizer.								
1	Realization of Logic gates							
2	4-bit ripple carry and carry look ahead adder using behavioral, dataflow and structural modeling							
3	Multiplexers (16:1, 4:1, 2:1 mux)							
4	3:8 decoder realization through 2:4 decoder							
5	8:3 Encoders							
6	Flip-Flops							
7	Synchronous and Asynchronous Counters							
8	Bi-directional shift register							
PART B (Any Five Experiments)								
Back-end Level Design and Implementation using EDA tools								
1.	Universal gates							
2.	An Inverter							
3.	Full Adder							
4.	Full Subtractor							
5.	D-Latch							
6.	2x4 Decoder							
EDA Tools/Hardware Required:								
1.	EDA Tool that supports FPGA programming including Xilinx Vivado.							
2.	Mentor Graphics Software / Cadence/Synopsys/Tanner or Equivalent Industry Standard/CAD Tool.							
3.	Desktop computer with appropriate Operating System that supports the EDA tools.							
Web Links:								
1.	https://courses.engr.illinois.edu/ece110/sp2021/content/courseNotes/files/?logicAndCMOS							
2.	http://ece-research.unm.edu/jimp/vlsiII/labs/layout_lecture.html							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS II								
Common to All Branches								
Part-A: Verbal and Soft Skills-II								
Course Objectives:								
1.	To expose the students to bettering sentence expressions and also forming equivalents.							
2.	To instill reading and analyzing techniques for better comprehension of written discourses							
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.							
4.	To inculcate conversational skills, nuances required when interacting in different situations							
5.	To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.							K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences							K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).							K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.							K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.							K3
SYLLABUS								
UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.							
UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis							

	on inferential and analytical questions), style and tone (to comprehend the author,,s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.
UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.
Reading/ Listening material:	
1.	Guide to IELTS, Cambridge University Press.
2.	Barron,,s GRE guide
3.	Newspapers like _The Hindu, _Times of India, _Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN
Text Books:	
1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.
Reference Books:	
1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie
5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.

Part-B: Quantitative Aptitude-II		
Course Objectives: The objective of introducing quantitative aptitude –II is:		
1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages	
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.	
3.	To familiarize the students with finding out accurate date and time related problems	
4.	To enable the students solve the puzzles using logical thinking.	
5.	To expose the students to various problems based on geometry and mensuration.	
Course Outcomes:		
S.No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3
SYLLABUS		
UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM, GM, HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.	
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood Relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.	
UNIT-III	Periods, Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.	
UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles	

UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi
3.	Puzzles to puzzle you by shakunataladevi
4.	More puzzles by shakunataladevi
5.	Puzzles by George summers
Reference Books:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3203	MC	--	--	3	0	--	--	--
ADVANCED CODING								
(Common to ECE & EEE)								
Course Objectives:								
1.	To understand the basics of Java Packages							
2	To learn about Collection Framework							
3	To investigate different methods of GUI Programming.							
4	To learn about Java collections and Libraries							
Course Outcomes: By the end of the course, the student should have the ability to:								
S.No	Outcome							Knowledge Level
1.	Able to solve problems using java collection framework and I/o classes.							K3
2.	Able to develop multithreaded applications with synchronization.							K3
3.	Able to develop applets for web applications							K3
4.	Able to design GUI based applications							K3
SYLLABUS								
UNIT-I (10 Hrs)	Standard Library templates and Java collections Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.							
UNIT-II (10 Hrs)	The Collections Framework (java.util)- Collections overview, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque. Accessing a Collection via an Iterator, Using an Iterator, The For-Each alternative, Map Interfaces and Classes, Comparators, Collection algorithms, Arrays, The Legacy Classes and Interfaces- Dictionary, Hashtable ,Properties, Stack, Vector.							
UNIT-III (10 Hrs)	More Utility classes, String Tokenizer, Bit Set, Date, Calendar, Random, Formatter, Scanner.							
UNIT-IV (8 Hrs)	GUI Programming with Swing – Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout. Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events.							
UNIT-V (12 Hrs)	Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, Applets – Applets and HTML, Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JCombo Box, Swing Menus, Dialogs.							

References:	
1.	Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd.
2.	Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education
3	An Introduction to programming and OO design using Java, J. Nino and F.A. Hosch, John Wiley & sons
4	Introduction to Java programming, Y. Daniel Liang, Pearson Education.
5	Object Oriented Programming through Java, P. Radha Krishna, University Press.
6	Programming in Java S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
7	Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.
8	https://www.geeksforgeeks.org/
9	https://www.tutorialspoint.com/